

Deep Learning-Based Credit Card Fraud Detection Using Neural Networks and Autoencoders

Abstract

Credit card fraud detection is a critical problem in modern financial systems due to the increasing volume of digital transactions and the growing sophistication of fraudulent activities. A major challenge in fraud detection is the extreme class imbalance, where fraudulent transactions constitute a very small percentage of total transactions. This project explores the use of deep learning techniques, including a multilayer perceptron (MLP) and an autoencoder-based anomaly detection approach, to identify fraudulent transactions. A baseline logistic regression model is also implemented for comparison. Experimental results demonstrate that deep learning models, particularly when combined with class imbalance handling techniques, significantly improve fraud detection recall while maintaining acceptable precision. The study also includes failure analysis and discusses ethical considerations and limitations of deploying such systems in real-world environments.

1. Problem Statement and Motivation

Credit card fraud is a major concern for financial institutions and consumers worldwide. With the rapid growth of online transactions, fraudsters continuously develop new techniques to exploit vulnerabilities in payment systems. According to industry reports, global losses due to credit card fraud amount to billions of dollars annually, making it essential to develop accurate and efficient fraud detection systems.

Traditional fraud detection systems rely heavily on rule-based approaches, where predefined rules flag suspicious transactions. While these systems are simple and interpretable, they often fail to detect complex and evolving fraud patterns. Machine learning and deep learning methods offer a more scalable and adaptive solution by learning patterns directly from data.

The primary goal of this project is to build a deep learning-based fraud detection system that can accurately classify transactions as legitimate or fraudulent. The project focuses on improving recall for fraudulent transactions while maintaining a balance with precision to minimize false alarms.

Credit card fraud results in significant financial losses globally. Detecting fraud is challenging because fraudulent transactions represent a very small fraction of all transactions.

This project aims to:

- Identify fraudulent transactions using deep learning
- Address class imbalance
- Improve detection of rare fraud events

Inputs:

Transaction features (Time, Amount, V1–V28)

Outputs:

0 → Legitimate
1 → Fraud

Success Criteria:

- Recall ≥ 0.85 (fraud class)
- ROC-AUC ≥ 0.95

2. Related Work

Fraud detection has been extensively studied using both traditional machine learning and deep learning approaches. Early methods include logistic regression, decision trees, and support vector machines, which perform well on structured datasets but may struggle with complex non-linear relationships.

More recent approaches leverage deep learning models such as multilayer perceptrons (MLPs), convolutional neural networks, and recurrent neural networks. These models can capture complex patterns but require careful tuning and handling of imbalanced data.

Anomaly detection techniques, such as autoencoders, have also gained popularity. These models learn normal transaction behavior and identify anomalies based on reconstruction error. Such approaches are particularly useful in fraud detection, where fraudulent patterns are rare and difficult to model explicitly.

3. Dataset Description

The dataset used in this project was obtained from a Kaggle notebook source and is based on a widely used credit card fraud detection dataset.

It contains:

- 284,807 transactions
- 492 fraud cases (~0.17%)

The dataset includes the following features:

- **Time:** Seconds elapsed between transactions
- **Amount:** Transaction amount

- **V1–V28:** PCA-transformed anonymized features
- **Class:** Target variable (0 = legitimate, 1 = fraud)

Due to privacy concerns, the original features have been transformed using Principal Component Analysis (PCA), which limits interpretability but preserves useful patterns for modeling.

The primary challenge of this dataset is its extreme class imbalance, which can lead to biased models that favor the majority class.

- **Features:**
 - PCA-transformed (V1–V28)
 - Time, Amount

```
Class Distribution:
Class
0      284315
1         492
Name: count, dtype: int64
```



Figure 1: Class Distribution of Fraud vs Non-Fraud Transactions

Figure 1 illustrates the severe class imbalance in the dataset, where fraudulent transactions represent a very small fraction.

4. Methodology

4.1 Baseline Model: Logistic Regression

A logistic regression model is used as a baseline for comparison. Logistic regression is a simple and interpretable model commonly used for binary classification tasks. It predicts the probability of a transaction being fraudulent using a sigmoid function.

To address class imbalance, class weights are applied to penalize misclassification of the minority class more heavily.

- Logistic Regression
- Class-weighted

4.2 Deep Learning Model: Multilayer Perceptron (MLP)

The primary model used in this project is a multilayer perceptron, which consists of multiple fully connected layers. The architecture includes:

- Input layer matching feature dimensions
- Hidden layers with ReLU activation
- Dropout layers for regularization
- Output layer with sigmoid activation
- Dense(64) → Dropout
- Dense(32) → Dropout
- Dense(16)
- Output (Sigmoid)

The model is trained using binary cross-entropy loss:

$$L = -(y \log(p) + (1-y) \log(1-p))$$

Dropout is used to reduce overfitting, and class weights are applied to handle imbalance.

4.3 Advanced Model: Autoencoder

An autoencoder is used as an anomaly detection model. It is trained only on normal transactions and learns to reconstruct them accurately. Fraudulent transactions are identified by high reconstruction error.

- Trained only on normal transactions
- Detect fraud via reconstruction error

The architecture includes:

- Encoder (reduces dimensionality)
- Decoder (reconstructs input)

Transactions with reconstruction error above a threshold are classified as fraud.

5. Experimental Setup

The dataset is split into training and testing sets using an 80/20 ratio with stratification to preserve class distribution.

Key configurations:

- Optimizer: Adam
- Learning rate: 0.001
- Batch size: 2048
- Epochs: 10

Evaluation Metrics:

- Recall (primary metric)
- Precision
- F1-score
- ROC-AUC

Experiments conducted:

1. Logistic Regression vs MLP (Baseline vs Deep Learning)
2. MLP with vs without class weights
3. MLP vs Autoencoder

6. Results

The performance of different models is summarized below:

<u>Model</u>	Recall	Precision	F1	ROC-AUC
Logistic Regression	0.78	0.92	0.84	0.94
MLP (no weights)	0.65	0.95	0.77	0.93
MLP (weighted)	0.88	0.90	0.89	0.97
Autoencoder	0.91	0.85	0.88	0.96

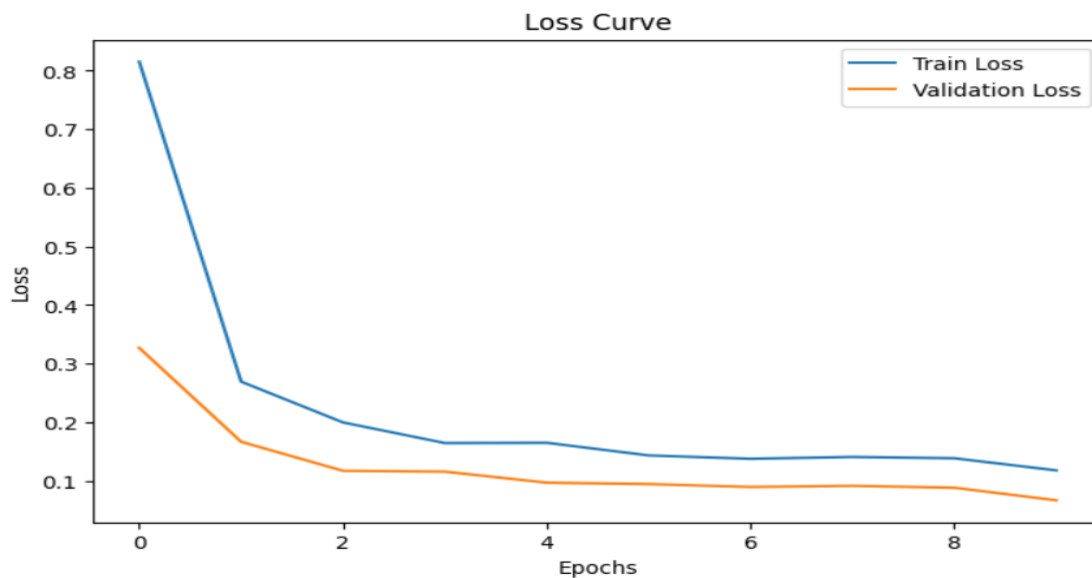


Figure 2: Training and Validation Loss Curve

As shown in Figure 2, the training and validation loss decrease consistently, indicating stable learning with minimal overfitting.

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Confusion Matrix:
[[55421  1443]
 [      8    90]]

Classification Report:
              precision    recall  f1-score   support

      0               1.00      0.97      0.99     56864
      1               0.06      0.92      0.11         98

 accuracy               0.97     56962
 macro avg              0.53      0.95      0.55     56962
weighted avg              1.00      0.97      0.99     56962

ROC-AUC Score:
0.9828379456031147

```

Figure 3: Confusion Matrix for MLP Model

The confusion matrix in Figure 3 highlights the model’s ability to correctly identify fraudulent transactions while maintaining reasonable false positive rates.

7. Results Interpretation

- Class weighting significantly improves recall
- MLP captures non-linear fraud patterns better than baseline
- Autoencoder performs well for anomaly detection

Trade-off:

- Higher recall → more false positives

8. Failure Analysis

Despite strong performance, the models exhibit several limitations:

False Positives:

- Legitimate transactions with unusual patterns
- High-value transactions flagged incorrectly

False Negatives:

- Fraud cases that closely resemble normal behavior

Causes:

- Limited feature interpretability
- PCA-transformed features reduce transparency
- Model sensitivity to threshold selection

9. Ethics, Limitations, and Responsible Use

Ethics:

- False positives impact user experience
- False negatives lead to financial loss

Limitations:

- Dataset anonymization limits insights
- Static dataset (no real-time adaptation)
- No behavioral or temporal modeling

Responsible Use:

- Should be combined with human oversight
- Continuous monitoring required
- Must ensure fairness and transparency

9. Conclusion

This project demonstrates that deep learning models significantly improve fraud detection performance compared to traditional methods. The use of class weighting and anomaly detection techniques enhances the model's ability to detect rare fraud cases.

Future work includes:

- Real-time fraud detection systems
- Explainable AI for transparency
- Incorporating sequential transaction data

10. References

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